Skills Summary

Multiplying by a One-Digit Number

Use this reference while completing your worksheet or when studying.

Skill 1 — The Standard Algorithm

Work right to left, one column at a time. Multiply the one-digit number by the ones, then the tens, then the hundreds, then the thousands.

When a column's product is 10 or more, write the ones digit in that column and **carry** the rest above the next column, then add it in after you multiply there.

A 0 digit still gets its turn: 0 times anything is 0, but a carry can land in that column.

Example: 47×6

Step 1. Both digits of 47 have to be multiplied, so I go column by column from the right and keep track of anything that spills over ten.

Step 2. Ones: $6 \times 7 = 42$, write 2 and carry 4 tens. Tens: $6 \times 4 = 24$ tens, plus the 4 carried tens is 28 tens, which is 280.

$47 \times 6 =$ **282**

Try it: $2{,}043 \times 8$ (watch the zero in the hundreds column)

check: **16344**

Skill 2 — Partial Products

Split the big number by place value, multiply each part, then add.

$1{,}325 = 1000 + 300 + 20 + 5$, so $1{,}325 \times 4$ is $4000 + 1200 + 80 + 20$.

Nothing is carried — every part keeps its own size — which makes this the method to fall back on when a carry goes wrong.

Example: 132×5 using partial products

Step 1. I want a method with no carrying to keep track of, so I break the number into its place values and multiply each piece on its own.

Step 2. $132 = 100 + 30 + 2$, so the parts are $5 \times 100 = 500$, $5 \times 30 = 150$ and $5 \times 2 = 10$; adding gives $500 + 150 + 10$.

$132 \times 5 =$ **660**

Try it: $2{,}140 \times 3$ using partial products

check: **6420**

Skill 3 — Estimate Before You Multiply

Round the big number, keep the one-digit number exact, multiply.

To round: look at the digit *one place to the right* of the place you are rounding to. If it is 5 or more, round up; otherwise round down.

The estimate tells you roughly how big the answer should be, so a product that is wildly bigger or smaller is a signal to check the work.

Example: Round 687 to the nearest hundred, ready to estimate 687×4.

Step 1. I only round the large number, because rounding the one-digit factor would throw the estimate away entirely.

Step 2. Rounding to the hundred means looking at the tens digit, 8; that is 5 or more, so 687 rounds up to the next hundred.

687 to the nearest hundred is **700**, so the estimate is $4 \times 700 = 2800$

Try it: Round 3,140 to the nearest thousand.

check: **3000**

(!) Watch out: never drop a 0 digit out of a number to make it shorter. Turning 5,704 into 574 removes an entire place value and shrinks the answer by thousands. The zero holds its column open — often for a carry.